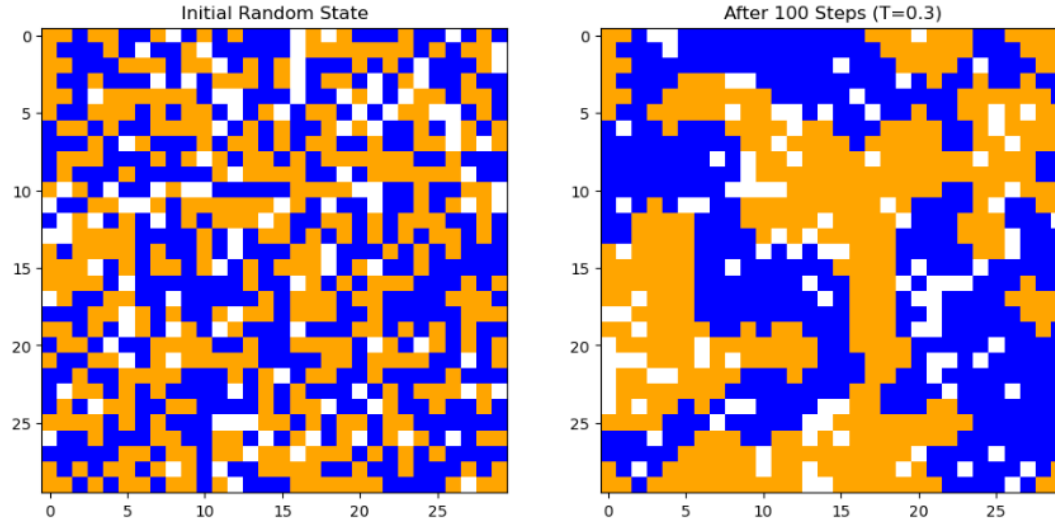

Extended Schelling Model

Location Preferences & Stochastic Relocation

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Original Schelling's Segregation Model



- Classic agent-based model
- Grid with colored agents (e.g. red and blue)
- Satisfaction threshold (e.g. at least 30% of neighbours similarity)
- Loop of: Check $\rightarrow$ Move $\rightarrow$ Repeat



Two Key Extensions

1. Location Preferences (α)

Agents not just care about their neighbor's type similarity but also have preference towards certain location (e.g. park, school, city center)

Segregation may form in the "desired location"

Revised Happiness Satisfaction Rule

$$H_i = \left(\frac{\text{Similar Neighbors}}{\text{Total Neighbors}} \right) + \alpha \cdot \frac{1}{1 + d_i}$$

d_i : Manhattan distance to nearest desirable area

α : Weight controlling location importance

2. Stochastic Relocation (p)

Agent now moves probabilistically, not deterministically. Even if it is not the best spot, they may relocate there. Simulate real-life scenarios (e.g. employment, family demands)

Disrupt emergent segregation patterns with the higher probability p of stochastic relocation



Problem & Motivation

Why Extend the Schelling Model?

To consider real-world scenarios of how individual choices emerge into large-scale segregation



Modelling Questions

- 01** How does introducing location preferences change the clustering dynamics shown in the original Schelling model?
- 02** What circumstances cause emergent segregation patterns to be disrupted or reinforced by stochastic relocation?





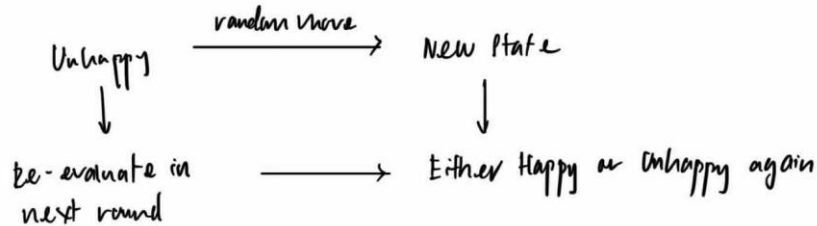
Modelling Techniques

Markov Chain Process

Discrete- time system that transitions from one state to another based only on the current state, not on any past states (memoryless)

Two state of the agents: Happy or Unhappy; Based on the proportion of neighbours of same type

State Diagram



Monte Carlo Simulation

Method to iteratively random sampling agent moves to compute results

Replicate the statistical behavior over a large number of iterations, agents and empty cells are updated randomly

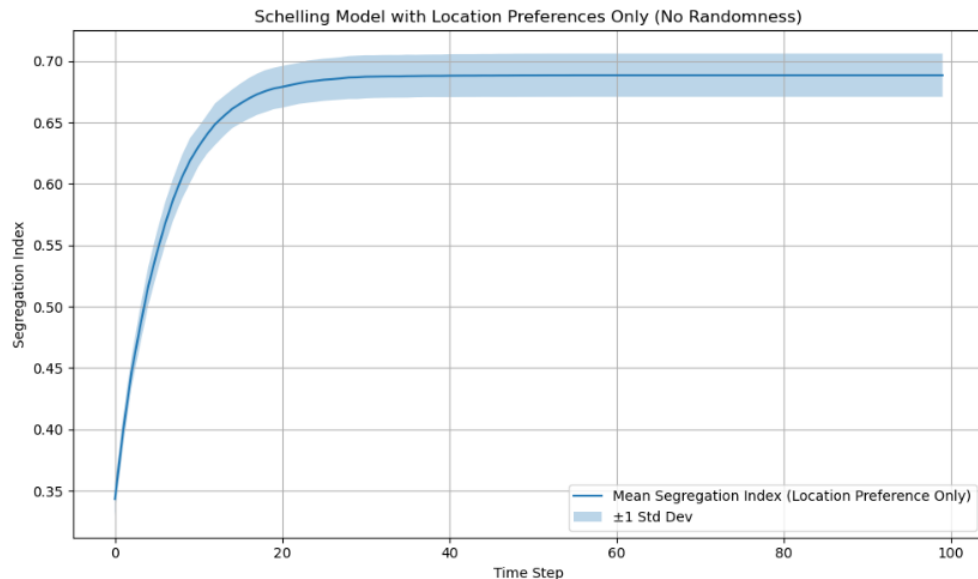
Evaluate steady-state segregation patterns under stochastic dynamics





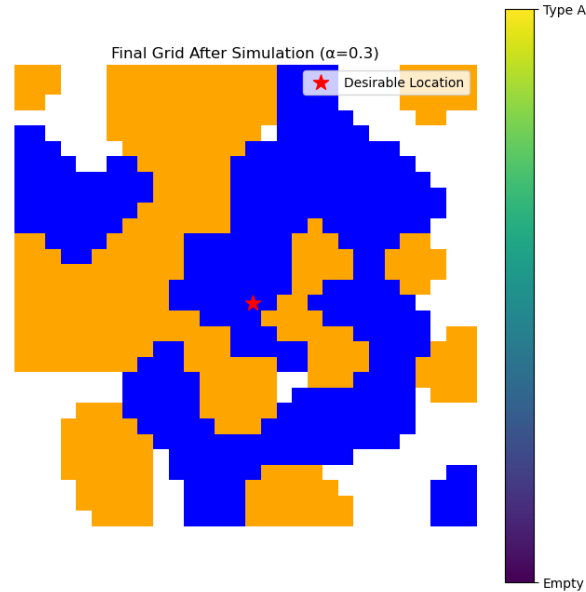
Results

Segregation Over Time Plot with Location Preferences (No Randomness)



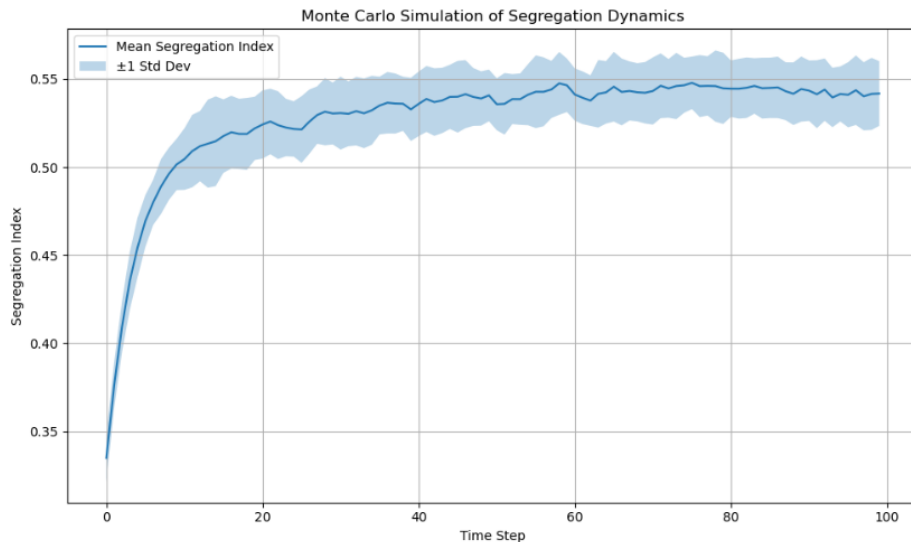
- Evaluates how similar an agent's neighbors are on average.
- Starts at ~0.34 (low segregation) and stabilises ~0.7 within 30 steps (high segregation)
- Deterministic location preferences lead to strong segregation and clustering even without randomness
- Narrow shaded region shows consistent behaviours during first 20 steps

Final Grid Visualisation (Without Randomness)



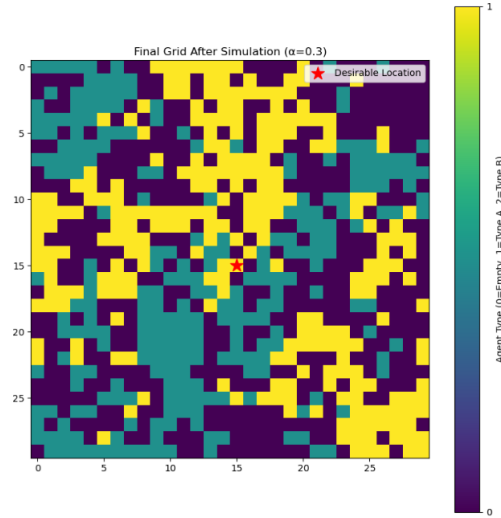
- Shows how the system stabilised over time
- Distinct clusters of Type A (blue) and Type B (orange) agents
- Higher agent density near the red star (desired location)
- Agent movement was influenced by distance to the desired location.
- Agents look for a desirable location, not only for the neighbours' similarity.

Segregation Over Time Plot with Location Preferences (With Randomness)



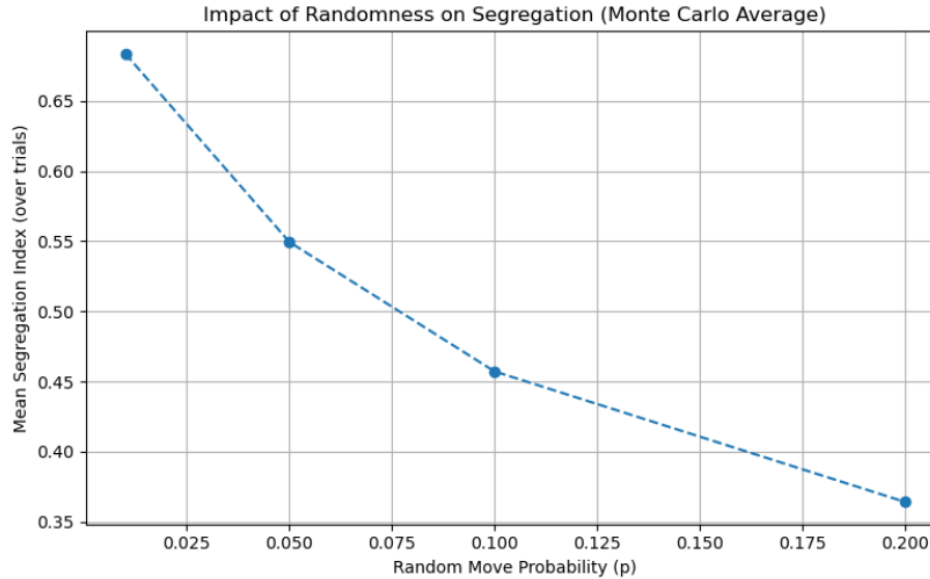
- Starts at ~0.33 and increases rapidly due to location preference (weight = 0.3) and neighbour similarity
- Stabilises ~0.55 after step 20
- Narrow Standard Deviation ± 1 shows a consistent result
- Full segregation is not achieved, indicating that randomness restricts extremes, and moderate clustering emerges.

Final Grid Visualisation (With Randomness)



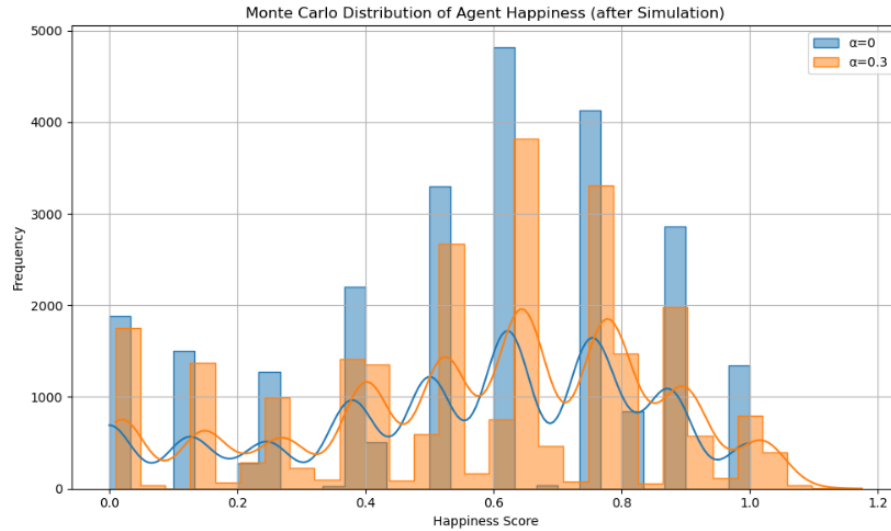
- Segregation is lessened but not eliminated by 5% random relocation and location preferences.
- Segregation stabilises at about 0.55
- Clustering is unaffected by low-level randomness; preferences still dominate
- Segregation remains stable because of strong neighbour and location preferences.
- Minor randomness aids agents avoid local suboptimal spots, subtly reinforce segregation

Randomness and Segregation Disruption



- Inverse Relationship: As randomness (p) increases, segregation decreases.
- Noticeable impact in disrupting segregation when $p \geq 0.10$
- The effect of low randomness ($p = 0.01$ or 0.05) is minimal.
- Random movement disrupts clusters when p is high enough

Happiness Distribution with/without Location Preferences



- The above histogram compares agent happiness for $\alpha = 0$ (blue) vs. $\alpha = 0.3$ (orange)
- At $\alpha = 0$: Happiness depends solely on neighbors which resulting more clustered distribution.
- Because of location preference, the curve moves to the right at $\alpha = 0.3$, resulting in increased overall happiness.
- Some agents remain unhappy ($H_i < 0.3$) if denied access to preferred desired locations.

Answering Modelling Questions

How does introducing location preferences change the clustering dynamics shown in the original Schelling model?

When location preferences are added, agents take into account both the neighbors' similarity and the distance to the desired location, which produces more realistic clustering around the chosen place



Answering Modelling Questions

What circumstances cause emergent segregation patterns to be disrupted or reinforced by stochastic relocation?

Segregation patterns are weakened by stochastic relocation when the random probability ($p > 0.1$). As higher randomness reduces overall segregation by disrupting clustering and reducing overall segregation



Thanks!

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